DEWDRIVE · PRODUCT REQUIREMENTS AND SPECIFICATIONS

DewDrive product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
1.1	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: PR-010 design wind a 25 m/s gust (item 4); fan cut-out 65 % RH in the assumptions (item 3)
1.0	2026-10-25	Amish Chadha	Kit 1.8 pipeline: DWD-REQ-001 migrated to DWD-PRS-001 (each R becomes a PR with the old number under Was, tolerance, units and the user requirement it traces to); PR-014 added for UR-009; status from DWD-CAL-001 v0.8 after the STEM check fixes and independent review (D16 to D20); PR-001 not met
0.9	2026-10-02	Amish Chadha	Status from DWD-CAL-001 v0.7 (script re-run; R9 met as restated); R11 over the target by $119 after the 130 °C glazing and the label
0.8	2026-10-02	Amish Chadha	R9 restated by Amish on 2026-10-02 for a lift with the trays and deck taken out, with an empty-before-lifting label (DWD-DEC-001, item 2); R9 met

0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.6	2026-10-01	Amish Chadha	Status from DWD-CAL-001 v0.4 for the constructable design (DWD-DDR-003); R9 and R11 not met, open for Amish
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish (R11 target $520; status from DWD-CAL-001 v0.3)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R7 restated for the drip screens; R8 and R10 wording; status from DWD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 update (design point and budget decided by Amish in DWD-DDR-001; R11 redefined to $500; status column from DWD-CAL-001)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with a design point and concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

DewDrive product requirements and specifications

What the design must do, each requirement traced to a user need in DWD-URS-001. This document replaces DWD-REQ-001 (v0.9); the old R number is kept under "Was". The requirements were checked by calculation at TRL 3 in DWD-CAL-001 v0.8, for the design decided by Amish on 2026-09-25 (DWD-DDR-002), made buildable in DWD-DDR-003 and corrected by the STEM check and its independent review on 2026-10-25 (DWD-DEC-001, D16 to D20): black absorber plates on the beds, PTFE-coated brine-wetted parts, an open tray floor of mesh on bars, a painted condenser underside and stronger ground anchors. On paper the design meets 10 of 14 requirements. **PR-001 is not met:** the design collects about 0.44 L per day at the design point against 0.5 L, and 0.37 to 0.41 L under the calculation's sensitivities (C9, C12, C13); the ways to recover it are for Amish (DWD-DEC-001, open item 2). PR-011 (cost) is over its value-engineering target, at an estimated $695 against $520. PR-006 and PR-010 can only be verified by test, and PR-005 is at risk if the fins shed less heat than the correlation says. Targets are proposals for review with a co-design partner, not user-validated needs. Method: STANDARDS section 22.

Design point

Decided by Amish on 2026-09-25 (DWD-DDR-001, D8). Unless a requirement says otherwise, it applies at this design point:

- Night: 10 h adsorption at 20 °C air and 40 % RH (6.92 g/m³ of vapour).
- Day: 6.0 kWh/m² of sun on the tilted aperture, of which 4.24 kWh/m² falls in the 09:00 to 15:00 window (DWD-CAL-001, C1); air at 35 °C.
- Low-humidity case: night air at 20 °C and 25 % RH, same sun.

Requirements

Table 1. Product requirements and specifications. Values and tags are from DWD-CAL-001 v0.8.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
PR-001	UR-001	Performance	Water at the design point	The unit shall collect water from night air at the design point	0.5	min	L/day	Analysis (DWD-CAL-001 cyclic balance), TRL 3; field test at TRL 4	Not met: 0.44 (C3); 0.37 to 0.41 under the sensitivities (C9, C12, C13)	R1
PR-002	UR-002, UR-001	Performance	Water in dry nights	The unit shall collect water when the night air is at 25 % RH	0.25	min	L/day	Analysis (DWD-CAL-001), TRL 3; field test at TRL 4	Met on paper: 0.31 (C4)	R2
PR-003	UR-008	Performance	Sun-only regeneration	The bed shall be regenerated by the sun alone and reach the target temperature by midday on a clear day at 35 °C	75	min	°C	Analysis (collector balance), TRL 3; logged at TRL 4	Met on paper: 105 °C at noon (C2)	R3
PR-004	UR-008	Electrical	Own electrical loads	The fan and logger shall use no more than the target energy per day, supplied by the on-board PV panel, with at least 2 days of battery autonomy	30	max	Wh/day	Analysis (energy budget), TRL 3; measurement at TRL 4	Met on paper: 23.6 against 35 Wh/day; 2.6 days (F1, F2)	R4
PR-005	UR-001	Performance	Condenser rise	The condenser shall stay within the target rise above ambient at the peak of desorption	15	max	K	Analysis (condenser balance, painted underside D18), TRL 3; logged at TRL 4	At risk: 11.9 K on the correlation (C2); about 14 to 16 K if the fins under the floor shed less heat (independent review)	R5
PR-006	UR-004	Materials	Water quality	Every wetted part shall be food-grade, and chloride in the	250	max	mg/L	Inspection (materials review), TRL 3; laboratory	To verify at TRL 4; materials	R6

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
				collected water shall stay below the target				water test at TRL 4	chosen to meet it	
PR-007	UR-012, UR-004	Safety	Salt containment	No $CaCl_2$ solution shall leave the tray and drip-screen assembly or reach the condenser after a night at 90 % RH; the pores shall stay below the target fill	100	max	% of pore volume	Analysis (pore and sump volumes, DWD-CAL-001 E), TRL 3; chamber test at TRL 4	Met on paper: 47 % after one night, 54 % after three with the fan rule; sumps 0.58 L (E2 to E6)	R7
PR-008	UR-003	Usability	Two actions per day	Daily operation shall need no more than two user actions (open the flaps at dusk, close them at dawn) taking the target time	5	max	min/ day	Analysis (task analysis), TRL 3; co-design sessions at TRL 4	Met by design	R8
PR-009	UR-007	Packaging and transport	Portable	The box, lifted with the trays and deck taken out, shall weigh no more than the target (two-person lift), carry an "Empty before lifting" label and be separable from the stand; two people shall set it up in 30 min or less. Restated by Amish on 2026-10-02 (DWD-DEC-001, item 2)	35	max	kg	Analysis (mass from model components), TRL 3; trial at TRL 4	Met on paper: 23.4 kg; 36.1 kg with everything inside; 61.1 kg in all (G1, G2)	R9
PR-010	UR-006	Physical architecture	Survive the site	The unit shall stay put in the target wind gust with two ground anchors (or ballast), use UV-stable glazing,	25	min	m/s	Analysis (wind load), TRL 3; supplier data and cycling test at TRL 4	To verify at TRL 4: two anchors of 234 N each at 25 m/s, specified 250 N, or 96	R10

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
				and keep the sorbent at 80 % or more of its capacity over 300 cycles; the target is a 25 m/s gust until the site value is known from the national wind annex (decided by Amish, 2026-10-25, DWD-DEC-001 item 4; was 20 m/s)					kg of ballast (H2 to H4)	
PR-011	UR-010	Physical architecture	Cost	The parts cost shall not exceed the value-engineering target in `project.yaml` (a hypothetical control target, not a limit)	520	max	USD	Analysis (priced BOM), TRL 3	Over target by $175: $695 (J1)	R11
PR-012	UR-011	Software	Record performance	The logger shall record air temperature and RH and the bed and condenser temperatures at the target interval for 30 days or more; the daily water mass is recorded by weighing	5	max	min	Analysis (storage estimate), TRL 3; demonstration at TRL 4	Met on paper: 8,640 records, 0.41 MB (I1)	R12
PR-013	UR-005	Safety	Protect users	Surfaces a person touches in normal use shall stay at or below the target; the glazing is labelled hot; no exposed conductor above 60 V DC	60	max	°C	Analysis (surface temperatures), TRL 3; logged at TRL 4	Met on paper: glazing outer skin 55 °C, fins 47 °C; 12.8 V DC (D1 to D3)	R13

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Wa
PR-014	UR-009	Physical architecture	Local build and repair	Every made part shall be made with hand tools, a drill, a rivet tool, a small press brake or folding bar and a coating shop, and every bought part shall come from a general supplier; no part needs special tooling	0	max	parts needing special tooling	Inspection (build plan DWD-BLD-001 and BOM), TRL 3; build at TRL 4	Met by design	non

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- The composite (25 wt % $CaCl_2$: 1.0 kg in 3.0 kg of mesoporous silica gel) is modelled from bulk salt thermodynamics: at equilibrium it holds 0.397 g/g at 40 % RH and 0.258 g/g at 25 % RH (DWD-CAL-001, A4). The yield is set by how fast the bed takes up and gives off water, not by these capacities. Measured isotherms and uptake rates of the actual composite must replace the model.
- The black absorber plates (DWD-DEC-001, D16) give an effective absorptance of about 0.89; the share of light passing the holes that the white beads and the plate underside absorb (0.70) is an estimate to measure at TRL 4 (DWD-CAL-001, C11).
- The day ends with the salt dried to about the dihydrate; the night starts with the bed at air temperature.
- With the fan stopped above 65 % RH (restarting below 60 %, DWD-DEC-001 item 3), natural airflow through the open flaps is taken as 4 m^3/h (10 % of the fan flow); this is an assumption to check by test.
- The tray floor's open share for the vapour (0.59: a mesh about 60 % open on bars 97.5 % open) is an estimate to confirm from the supplier's mesh data (DWD-DEC-001, D19).
- One unit is a supplement: about 0.44 L per day is about one sixth of one person's survival drinking need of 2.5 to 3 L per day (WHO technical note 9).

> **Safety:** PR-006, PR-007, PR-010 and PR-013 exist because of hazards: untreated water, a corrosive salt solution, wind loads on a 1.1 m^2 panel, and hot surfaces. They cannot be waived without a documented decision by Amish.